Lecture 1: Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science that builds systems able to perform tasks that normally need human intelligence, such as understanding language, recognizing images and making decisions.

Types of AI:
- Narrow AI: designed for one task, for example a spam filter or a chess program.
- General AI: a hypothetical system that could perform any intellectual task a human can do.

Machine Learning (ML) is a subfield of AI where computers learn patterns from data instead of being programmed with fixed rules. The three main types of machine learning are supervised learning, unsupervised learning and reinforcement learning.

In supervised learning the model is trained on labelled examples, such as emails marked as spam or not spam. In unsupervised learning the model finds structure in unlabelled data, for example grouping customers into clusters. In reinforcement learning an agent learns by trial and error, receiving rewards for good actions.

Lecture 2: Neural Networks

A neural network is a model inspired by the human brain. It is made of layers of connected units called neurons. Each connection has a weight, and each neuron applies an activation function such as ReLU or sigmoid.

A typical network has an input layer, one or more hidden layers, and an output layer. Networks with many hidden layers are called deep neural networks, and training them is called deep learning.

How a neural network learns:
1. Forward pass: the input moves through the layers and the network makes a prediction.
2. Loss function: the error between the prediction and the correct answer is measured.
3. Backpropagation: the error is sent backwards to calculate how much each weight contributed to it.
4. Gradient descent: every weight is updated a little in the direction that reduces the loss.

This cycle is repeated for many epochs until the loss becomes small. The learning rate controls how big each update step is.

Lecture 3: Natural Language Processing and Embeddings

Natural Language Processing (NLP) helps computers understand human language. Common NLP tasks are translation, sentiment analysis, question answering and text summarization.

Computers cannot understand words directly, so text is converted into numbers. An embedding is a list of numbers (a vector) that represents the meaning of a word or a sentence. Sentences with similar meaning have vectors that are close to each other.

The similarity between two embeddings is often measured with cosine similarity. A value close to 1 means the meanings are very similar.

Transformers are the modern architecture for NLP. They use a mechanism called attention, which lets the model focus on the most relevant words in a sentence. Large Language Models (LLMs) such as Gemini and Claude are built on transformers and trained on huge amounts of text.

Lecture 4: Retrieval-Augmented Generation (RAG)

LLMs sometimes produce answers that sound correct but are wrong. This problem is called hallucination. Retrieval-Augmented Generation (RAG) reduces hallucination by giving the model real documents to read before answering.

RAG pipeline:
1. Split documents into small chunks.
2. Convert each chunk into an embedding.
3. Store the embeddings in a vector database such as ChromaDB, FAISS or Pinecone.
4. When a user asks a question, convert the question into an embedding.
5. Retrieve the most similar chunks (top-k) from the vector database.
6. Send the question together with the retrieved chunks to the LLM, which writes the final answer.

A vector database is different from a normal SQL database. A SQL database finds rows by exact values, while a vector database finds items by similarity of meaning. Semantic search is search based on meaning rather than exact keywords.

Lecture 5: Evaluation and Ethics

Machine learning models are evaluated on data they have not seen during training. The dataset is usually split into a training set, a validation set and a test set.

Common evaluation metrics:
- Accuracy: the percentage of correct predictions.
- Precision: of all items predicted positive, how many were really positive.
- Recall: of all real positive items, how many the model found.
- F1-score: the harmonic mean of precision and recall.

Overfitting happens when a model memorizes the training data and performs poorly on new data. Regularization, dropout and more training data help to reduce overfitting.

AI ethics is also important. AI systems can show bias if their training data is unfair. Developers should protect user privacy, explain how their systems make decisions, and make sure AI is used responsibly.